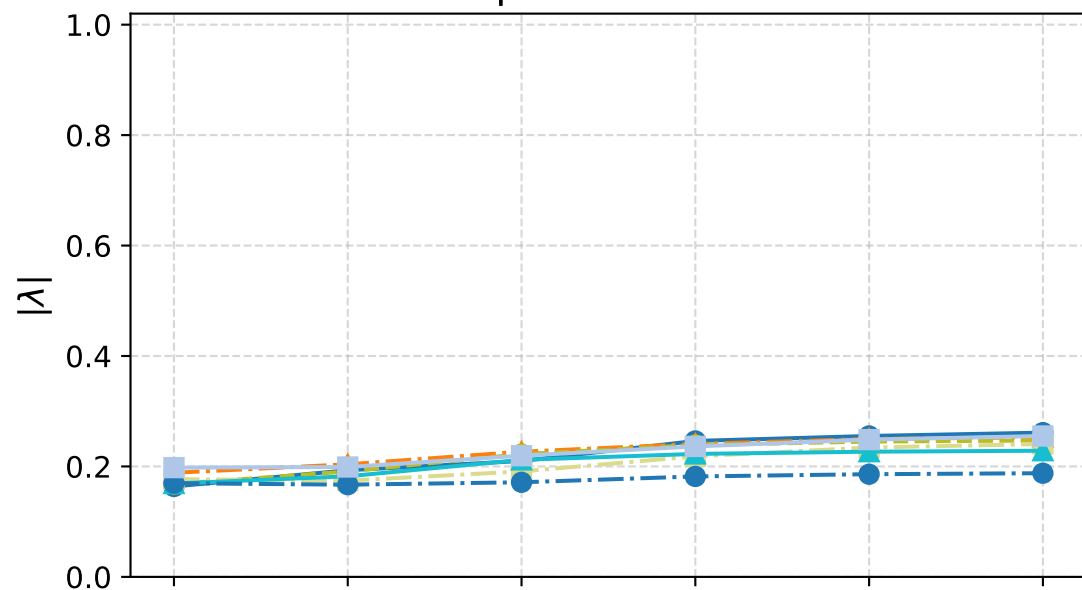


HCurl Laplacian: W-Cycle (Row Sum)

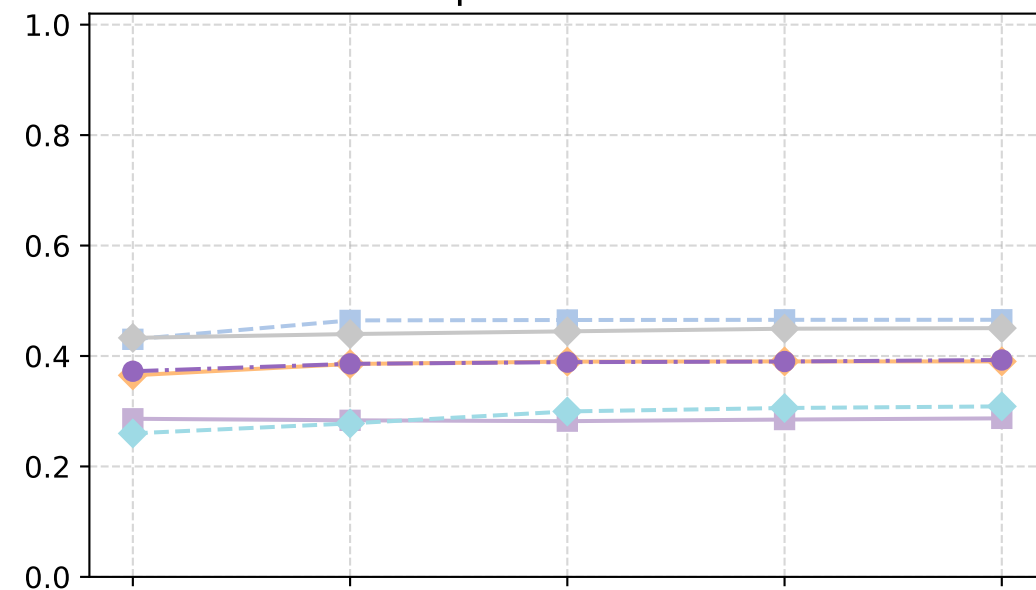
2D

3D

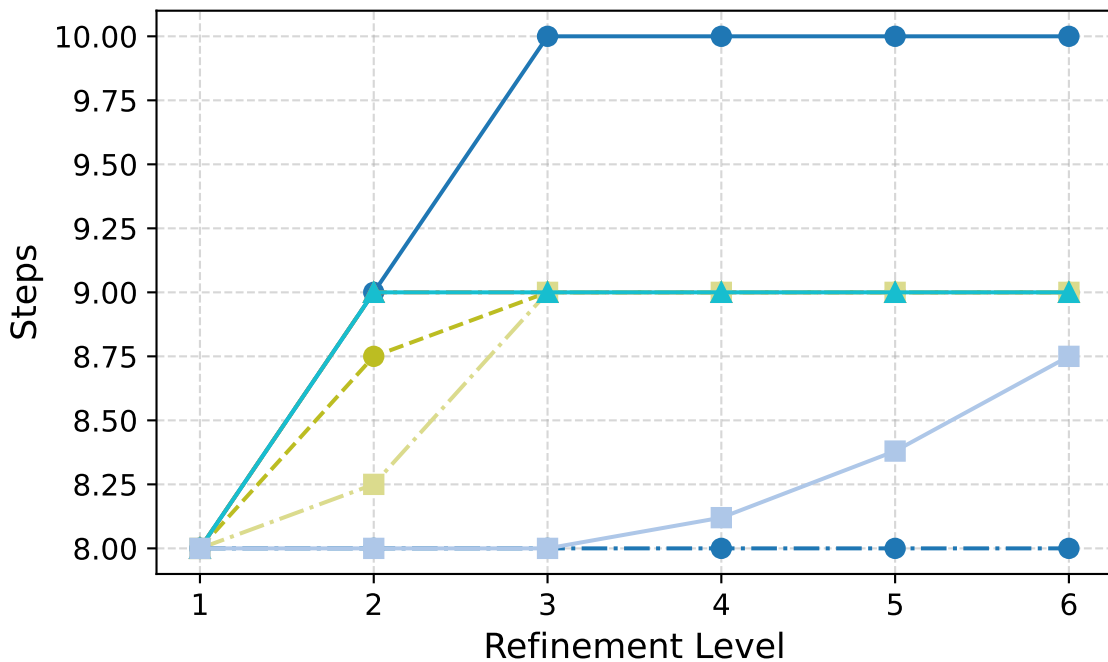
Spectral Radii



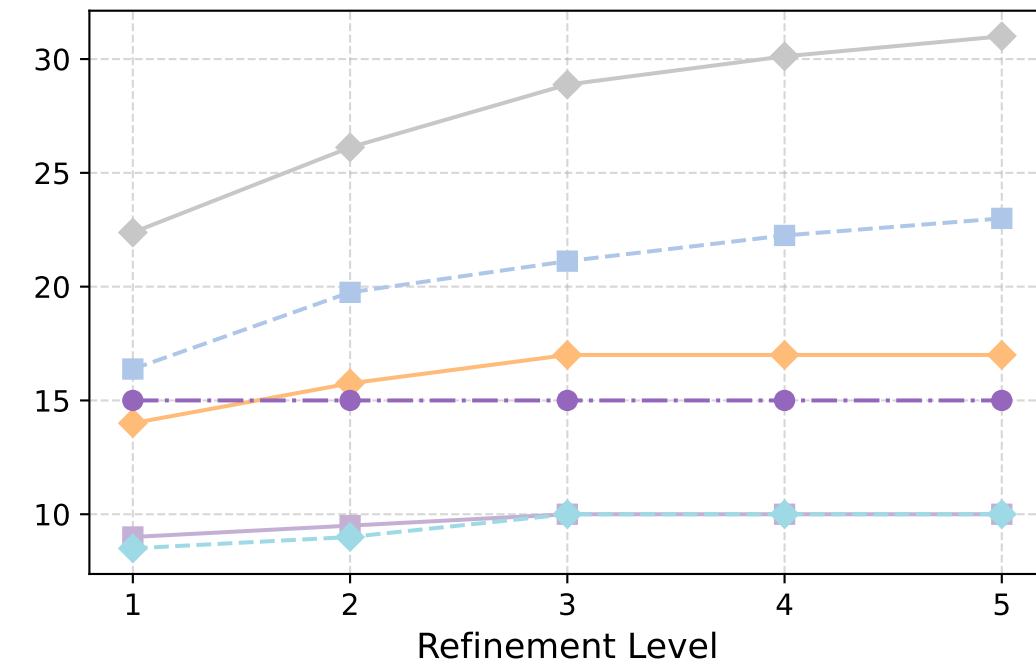
Spectral Radii



GMRES Iterations



GMRES Iterations



- circle ($n_{ev} = 1$)
- L ($n_{ev} = 1$)
- reentrant ($n_{ev} = 1$)
- square_centered ($n_{ev} = 1$)
- square_hole ($n_{ev} = 2$)
- tria ($n_{ev} = 1$)
- two_holes ($n_{ev} = 3$)

- ball ($n_{ev} = 1$)
- corner ($n_{ev} = 1$)
- corner_structured ($n_{ev} = 1$)
- cube ($n_{ev} = 1$)
- hole_void ($n_{ev} = 2$)
- tetra ($n_{ev} = 1$)